

Why Your Backtest Is Lying To You

40 Experiments, One Honest Postmortem — The FinPilot Research Handbook

FinPilot Research

2026

Contents

Introduction: Why We Wrote This Down	2
Chapter 1 — The Setup: What We Thought We Had	3
Chapter 2 — Mistake #1: Measuring the Wrong Thing	4
Chapter 3 — Mistake #2: Row Count Is Not Sample Size	5
Chapter 4 — Mistake #3: Chasing a Signal Without Trying to Kill It First	6
Chapter 5 — Mistake #4: “Good Detector” Is Not “Good Selector”	7
Chapter 6 — Mistake #5: The Silent Duplicate-Row Trap	8
Chapter 7 — Mistake #6: Marking a Finding “Closed” Before It Clears Its Own Bar	9
Chapter 8 — What Actually Survived	10
Chapter 9 — A Protocol You Can Copy	11
Chapter 10 — Closing: “We Don’t Know” Is a Feature	12
Appendix A — Quick-Reference Checklist	13
Appendix B — Glossary	13

Introduction: Why We Wrote This Down

FinPilot is a US-equity momentum scanner. Over roughly two years we built a scoring model, an eligibility filter (`entry_ok`), a conviction tier system, and a research pipeline to back it all up. We ran forty-plus experiments against it in the summer of 2026. This handbook is the honest result.

We are not writing this because we found an edge and want to teach you how we did it. We are writing this because we spent two years believing we might have one, and a disciplined audit — the kind almost nobody runs on their own system — showed that we didn't. What we found instead was more useful than an edge: a repeatable list of the exact ways a backtest lies to the person who built it, and a protocol that catches each one before it reaches a live decision.

If you are building, backtesting, or buying into a trading signal, a scanner, or a “smart money” scoring system, every mistake in this handbook is one you are at risk of making right now, quietly, without anything telling you it happened. That's what happened to us. Nothing crashed. No error message appeared. The system ran for months producing numbers that looked like evidence. They weren't.

This is a field guide to finding out before it costs you more than our two years cost us.

Chapter 1 — The Setup: What We Thought We Had

FinPilot scans the US equity universe several times a day, scores every candidate on a composite of momentum, volume, gap, and volatility features, and marks a subset as `entry_ok` — eligible for a trade idea. A secondary `conviction_tier` (A/B/C) ranks eligible names further. Every one of these mechanisms was tuned, over roughly two years, against a backtest that showed the system’s picks outperforming.

Here is the composite score’s actual feature list, unremarkable to anyone who has built a scanner before:

- `dist_52w_high` — distance from the 52-week high
- `past_5d_pct` — 5-day trailing return
- `lottery_factor`, `overnight_gap_factor` — gap/volatility-shape proxies
- `rvol`, `atr_pct_real` — relative volume, volatility
- `catalyst_factor`, `sentiment` — intended news/catalyst signal
- `squeeze_factor` — short-squeeze proxy

The scanner’s own backtest, `resolved_pct_t5`, reported a 5-day forward outcome for every signal. It looked healthy. Conviction tier A looked better than B, B better than C. The story hung together.

None of it was checked against a genuinely independent, cost-adjusted, day-clustered, null-controlled test until this summer. Once it was, the story did not survive contact with its own data.

Chapter 2 — Mistake #1: Measuring the Wrong Thing

The bug: `resolved_pct_t5` was computed as:

```
resolved_pct_t5 = (max(high[T+1..T+5]) - entry_price) / entry_price * 100
```

That is not a return. It is the **Maximum Favorable Excursion (MFE)** — the best price the stock touched at any point in the following five sessions, whether or not you were still holding when it happened, whether or not it closed anywhere near there. A stock that spikes to +9% on day 2 and closes at -4% on day 5 registers as “+9%” under this field. Every backtest, every score-calibration check, every conviction-tier ranking that used `resolved_pct_t5` was quietly grading the system on the best five minutes of a five-day window, not on what actually happened to someone holding the position.

How we found it: not by reading the code first — by an inconsistency between two supposedly-related metrics. Average True Range (ATR) correlated at **+0.40** with `resolved_pct_t5`, but only **+0.01** with the honestly-recomputed close-to-close return over the same window. A real “volatility predicts outcome” relationship should show up in both. It only showed up in the metric that was secretly measuring best-case excursion — because *of course* higher-ATR names touch a higher intraday high. The “edge” was measuring the wrong thing, and the mismatch between two versions of the same claim is what exposed it.

The fix added three honestly-labeled fields — `c2c_1d`, `c2c_5d` (realized close-to-close return) and `mae_t5` (maximum adverse excursion, the same trap in the other direction) — and re-ran the core comparison. Under the honest label, the `entry_ok` cohort’s picture changed materially versus the MFE-based version.

The second mistake, layered on the first: when we re-ran this fix, the first report we produced framed the reversal as absolute and immediate — “one label fix overturned two years of edge.” We tested that specific claim the same way we tested everything else in this handbook (see Chapter 4) and it did not survive day-level clustering — the point estimate stayed in the same direction, but it was not statistically decisive on its own. The label bug was real and the fix was necessary. The *dramatic one-sentence version of what the fix proved* was oversold, by us, in our own internal report, within hours of finding the bug. We caught it, corrected the write-up, and logged the correction next to the original claim. We are telling you about this specifically because “we just fixed a real bug” is exactly the moment you are most tempted to skip the next four chapters of scrutiny on whatever the fix seems to show. We nearly did.

The lesson: audit what your labels actually measure before you audit your model. A model trained or graded against MFE-as-return will always look better than it is, in direct proportion to how volatile the underlying names are — which means it will look *best* exactly where it is *most* dangerous.

Chapter 3 — Mistake #2: Row Count Is Not Sample Size

We had 53,859 raw rows in our canonical research export. After removing duplicate scans of the same symbol on the same day (our scanner runs hourly, and a name can be re-scored many times before the market closes — more on this in Chapter 6), we had 27,361 unique symbol-day observations. That still sounds like a lot.

It isn't, for the purpose of judging whether a result is real. Two forces shrink it:

1. **Cross-sectional correlation.** All the symbols scanned on the same day share the same market move. Fifty rows from one Tuesday are not fifty independent trials — they are one trial with fifty names attached to it.
2. **Serial correlation.** Yesterday's market outcome is a noisy predictor of today's. Day-level means are themselves autocorrelated (we measured a lag-1 autocorrelation of 0.23 on daily mean returns), which shrinks the number of *independent* days below the raw day count too.

Once we ran a date-block bootstrap to estimate genuine effective sample size, the number came back at roughly **620 effective observations** for the full universe — about **1/44th** of the raw row count — and roughly **168** for the `entry_ok` eligible cohort specifically, against a nominal count in the high hundreds.

Every confidence interval, every “statistically significant” claim we or anyone else had produced against this dataset using naive row-count-based standard errors was overstating precision by roughly that same factor. A result that looked comfortably significant at $n=27,361$ could be, and in several cases was, well within noise at $n \sim 620$.

The lesson: before you trust a p-value or a confidence interval on cross-sectional financial data, ask what the *effective* sample size is — not the row count. If nobody has computed it, assume it is smaller than you think by an order of magnitude, because it usually is.

Chapter 4 — Mistake #3: Chasing a Signal Without Trying to Kill It First

This is the chapter we are proudest of, because it documents us being wrong twice on the same finding before we let it go.

Round 1 — row-level. We tested reverse-ranking the composite score: buying the bottom 20% by score instead of the top 20%, on the theory that a broken score might be broken in a *consistent* direction. On a naive, row-level split into an in-sample and out-of-sample half, this looked like the strongest finding in the program — the bottom-20% cohort beat the top-20% cohort in both halves.

Round 2 — day-clustered. Applying the day-clustering standard from Chapter 3, the in-sample half stayed marginally positive ($t \sim +2.2$) but the out-of-sample half collapsed ($t \sim -0.15$) — a handful of extreme days were driving the entire out-of-sample result. Not dead yet, but no longer “the strongest finding in the program.”

Round 3 — deduplicated, quarterly, matched-random. We deduplicated the underlying rows properly (Chapter 6), split the period into four quarters instead of one arbitrary in-sample/out-of-sample cut, and added a genuine matched-random control — not just “the rest of the universe,” but a same-day, same-size random draw. The result: the effect appeared in one quarter out of four, was confounded with a bad quarter for the *top*-ranked cohort (not a good quarter for the bottom) in a second, and was flat-to-noise in the remaining two. The full-period aggregate against the random control was not significant.

Verdict: closed. We logged it as an artifact, not a finding, and moved on.

We call the four checks above the **artifact ladder**. A cross-sectional financial result should climb every rung before you act on it:

Rung	Question
1. Row-level split	Does it appear at all, in-sample vs. out-of-sample?
2. Day-clustered	Does it survive when same-day correlation is priced in?
3. Deduplicated, blocked	Does it survive across <i>multiple</i> independent time blocks, not one arbitrary cut?
4. Matched-random control	Does it beat a random draw of the same size, same day, same universe — or did the <i>other</i> side just have a bad stretch?

A result that only clears rung 1 is not a finding. It’s a discovery signal, and it should be labeled that way in every report until it clears rung 4.

Chapter 5 — Mistake #4: “Good Detector” Is Not “Good Selector”

This is the finding that hurt the most, because it is the one closest to the actual product. Three independent tests, using three different comparison groups, agreed:

Test 1 — vs. random rejection. On any given scan day, the `entry_ok`-eligible portfolio underperformed a same-day, same-size random draw from the *rejected* names. Median gap: **−2.01 percentage points**. The eligible cohort was ahead of the random-rejected control on only **31% of trading days**.

Test 2 — inside the same score band. This is the one that ruled out “the score just needs a better threshold.” Restricting the comparison to *only* names in the single best score quintile — apples-to-apples on score — the names `entry_ok` additionally selected still underperformed the ones it rejected *from the same band*: median **−0.20%** for eligible vs. **+1.08%** for the excluded names in the identical score quintile. Whatever `entry_ok` was doing beyond the score, it was making things worse, not better.

Test 3 — vs. benchmark. Against SPY over the same holding window, the eligible cohort’s median relative return was **−1.22 percentage points**, with a bootstrap confidence interval entirely below zero.

Three different counterfactuals, three different confirmations of the same direction. This is what “the selection layer subtracts value” looks like when you insist on more than one control group.

Why does the score correlate with anything if it carries no forward information? Because it correlates strongly with *the recent past*, not the future. The score’s two largest components — distance from the 52-week high ($\rho = 0.667$) and the trailing 5-day return ($\rho = 0.376$) — are both backward-looking. The two features carrying the most (still small, but real) *forward* information in the whole feature set were different ones entirely, and pointed the opposite direction the score was weighting them: `lottery_factor` ($\rho = -0.110$) and `overnight_gap_factor` ($\rho = -0.095$).

Put plainly: the score is a well-built detector of *what already happened*. It is not, currently, a predictor of what happens next — and the one place a faint, real, forward-looking signal exists in the data, the score is weighting it away from, not toward, where the evidence points.

The lesson: if your selection layer’s job is to add value on top of your score, test it against a matched random control from the *same population*, not against “the strategy vs. doing nothing.” A layer can beat doing nothing and still actively destroy value relative to chance. Ours did.

Chapter 6 — Mistake #5: The Silent Duplicate-Row Trap

Our scanner runs hourly. A liquid, active name can be re-scored ten or more times in one session as its price, volume, and gap features update intraday. Each scan writes a new row.

That’s reasonable behavior for a live product. It is a landmine for research, if the research pipeline treats every row as an independent observation. We found that of 53,754 total rows in one export, only 27,323 represented unique (`symbol`, `date`) pairs — a **48% duplication rate**, with one name scanned seventeen times in a single day at the extreme.

The direct statistical harm was limited — duplicate-count did not correlate strongly with the score or the outcome (rank correlation around 0.03) — but the *indirect* harm was real: every day-level average, every “n=53,754” quoted anywhere in a report, was silently weighting frequently-rescanned names more heavily than the rest of the universe, and every sample-size claim was inflated by roughly double.

The lesson: if your data pipeline can emit more than one row per real-world decision point, deduplicate to one canonical row — earliest timestamp, latest timestamp, whichever matches when the decision was actually made — before any aggregate statistic touches the data. Do this before Chapter 3’s day-clustering, not after; day-clustering on duplicated rows still overweights the same names within each day’s cluster.

Chapter 7 — Mistake #6: Marking a Finding “Closed” Before It Clears Its Own Bar

Even after we wrote the artifact ladder in Chapter 4, we broke our own rule once more — on the very fix from Chapter 2.

We had proposed a mechanism to explain why `entry_ok` might select adversely: an “extension/exhaustion” story, where the eligibility filter systematically picked names that had already made their move (high `dist_52w_high`, high recent return), which should mean-revert. The initial evidence looked strong: sorting by an ATR-normalized extension measure, the `entry_ok` rate rose roughly **15-fold** from the lowest to the highest decile. We wrote this up as the program’s first concrete, causal explanation for the selection-layer problem.

It was descriptive, not causal, and we should have run it through the artifact ladder before calling it a diagnosis. When we did — testing whether extension *itself* predicted worse forward returns in a clean, day-clustered, full-population test, and whether capping eligible names by extension level actually recovered performance — neither held up. The decile-vs-return relationship was not monotone. Capping extension on the eligible cohort produced no significant improvement over the uncapped cohort, in either direction. We downgraded our own “diagnosis” back to “hypothesis” and logged the correction next to the original claim, the same day.

The lesson: the artifact ladder in Chapter 4 applies to *causal explanations* for a finding just as much as it applies to the original finding. “We found a plausible mechanism” is a discovery signal. It graduates to a finding only after it survives the same scrutiny you’d apply to any other claim — including the ones you’re personally attached to because you thought of them.

Chapter 8 — What Actually Survived

None of this means nothing worked. Three things held up under every test we could throw at them, and none of the three require forecasting a stock's direction:

Volatility-to-adverse-move relationship. ATR predicted the magnitude of adverse excursion (maximum drawdown within the holding window) with an information coefficient around -0.51 , holding across regime splits and surviving a look-ahead audit of the underlying calculation. This is a genuine, reproducible risk relationship — it does not tell you what to buy, but it tells you how much room a position needs.

Concentration limits. An unconstrained top-10-by-score daily portfolio put an average of **62% of its exposure in a single sector** on a typical day — pure accident of an unconstrained ranking, not a sector call anyone made on purpose. Simply capping exposure at three names per sector — no forecasting skill required — cut daily return volatility and 5%-tail risk (CVaR) by roughly half, cut maximum drawdown by roughly half, and *raised* average return, in the same backtest window.

Volatility-parity position sizing. Sizing positions inversely to volatility instead of equally weighting them produced a materially better drawdown profile (-15.9% vs. -24.3% max drawdown in one construction test) and the best risk-adjusted daily return of the schemes tested — again, with no change to which names were selected.

The pattern across all three: **construction and risk management were the reliable value-add. Selection was not.** If you are building a similar system and you are short on time, this is the order in which to trust your own results by default — tighten construction and sizing before you tighten the picker.

Chapter 9 — A Protocol You Can Copy

We turned the six mistakes above into a four-gate research protocol. The rule is simple: a claim may not be called a “finding” until every gate below it is closed. While a gate is open, anything produced above it is labeled a **discovery signal** — worth writing down, not worth acting on.

Gate 1 — Data. Is every label’s actual definition verified against the source code that produces it (not assumed from its name)? Is price history free of unflagged discontinuities? Is it known, for every feature, at what point in time it was actually knowable?

Gate 2 — Measurement. Is the *effective* sample size reported, not the row count? Does every new result carry a matched-null or permutation-based comparison, run *before* the result is written up, not after it looks interesting? Is the total number of hypotheses and configurations tested — the multiple-testing budget — tracked anywhere?

Gate 3 — Execution. Is there any evidence the signal survives real-world frictions — spread, fill delay, capacity — or only a frictionless daily-bar simulation? How long does the signal’s predictive power actually last before it decays?

Gate 4 — Signal. Only once 1–3 are closed does a specific predictive or selection claim get tested at all, and even then, only against a pre-registered hypothesis and a clean holdout that was not used to generate the idea.

If you build only one artifact from this handbook, build a one-page version of this gate table for your own project, and refuse to write the word “edge” or “finding” in any report until you can point to which gate a given result has cleared.

Chapter 10 — Closing: “We Don’t Know” Is a Feature

Every claim in this handbook is checkable against our own published research log — we did not summarize our failures into something flattering. Where we overstated a result to ourselves, we said so, in the same document, on the same day, next to the original claim.

We think this is the actual differentiator worth building around, more than any score or signal: a product that tells you honestly what it has and has not proven, in a market where almost nothing else does. That’s the direction we’re taking FinPilot next — not a better prediction, but a more honest one, with the construction and risk-management layers (Chapter 8) doing the load-bearing work while the selection layer stays frozen until it can pass its own gates.

If any part of this matched a system you’re building or evaluating, we’d like to hear about it — and if you want a second pair of eyes running your own results through the four-gate protocol before you trust them, that’s a conversation worth having.

FinPilot Research — 2026

Appendix A — Quick-Reference Checklist

Before you call anything a finding, answer all seven:

1. Is the label I’m measuring actually what its name says it is — verified in code, not assumed?
2. What is the *effective* sample size, not the row count?
3. Did I deduplicate to one row per real decision point before aggregating?
4. Does this result survive day-level (or equivalent cluster-level) clustering?
5. Does it hold across multiple independent time blocks, not one in-sample/out-of-sample cut?
6. Does it beat a matched random control from the same population — not just “doing nothing”?
7. If I have a causal story for *why* it’s true, has that story itself cleared checks 4–6?

If you answered “no” or “not sure” to any of these, you have a discovery signal. Write it down. Don’t trade it, price it, or publish it as proven yet.

Appendix B — Glossary

Terms coined or sharpened over the course of this postmortem:

Effective sample size — the number of statistically independent observations actually present in a dataset, after accounting for correlation between rows (e.g. same-day, same-symbol). Almost always smaller than the row count in financial panel data.

MFE / MAE (Maximum Favorable / Adverse Excursion) — the best/worst price a position touches within a holding window, regardless of entry or exit timing. Not a realized return.

Day-clustering — treating all observations from the same calendar day as correlated with each other when computing standard errors, rather than treating every row as independent.

Matched-random control — a random sample drawn from the same population, on the same days, at the same size, as the group being evaluated — the correct baseline for testing whether a selection rule adds value.

Discovery signal vs. finding — a discovery signal is a result that has not yet cleared a pre-registered, null-controlled, out-of-sample test; a finding has. The distinction should appear in the label of every result, not just in a footnote.

Artifact ladder — the sequence of four checks (row-level, day-clustered, multi-block, matched-random) a cross-sectional result should pass before being trusted.

Terms every reader of a scanner, screener, or signal product should already know before trusting one — reproduced from FinPilot’s own public glossary:

Calibration — A probability is calibrated when it matches reality: a signal that says “60%” should come true about 60% of the time it fires, over many trials. Most retail signal products never publish this check.

Base Rate — how often an outcome happens with *no* filtering applied at all. Every signal, screener, or scanner should be judged against this baseline, not against zero.

Lift — a signal’s hit rate divided by the base rate. A lift above 1.0 means the filter is genuinely adding something beyond chance; at or below 1.0, as in the `entry_ok` case in Chapter 5, the filter is subtracting value even while it “looks” selective.

Out-of-Sample Test — testing a rule on data it was not tuned on. The real exam; a polished in-sample backtest is not one.

Walk-Forward Validation — repeatedly training on the past and testing on the next period, rolling forward through time, rather than one static train/test split.

Survivorship Bias — judging a strategy only on names that still exist today; ignoring delisted, bankrupt, or acquired names makes historical results look systematically better than they were.

Drawdown — the decline from a portfolio's peak value to its subsequent trough. The core measure of how much pain a strategy inflicts on the way to any eventual gain — and, per Chapter 8, the one place this program's evidence was most robust.

Position Sizing — deciding how much capital a single idea deserves. Independent of whether the idea itself is good, this is where Chapter 8's strongest results lived.

Risk/Reward Ratio — potential gain versus potential loss on a trade. A favorable ratio still needs a realistic *probability* behind it — a great risk/reward on a coin-flip is still a coin-flip.

Relative Volume (RVOL) — a stock's current volume relative to its own normal level. High RVOL is a real, observable fact about a stock's day; it is not, by itself, evidence of a repeatable edge — see Chapter 5.

Post-Earnings Drift (PEAD) — the well-documented tendency for a stock to keep drifting in the direction of a strong earnings surprise for days afterward — one of the few genuinely studied, peer-reviewed anomalies in this space, and a useful sanity check for what a *real*, published forward-looking effect size actually looks like next to the near-zero correlations in Chapter 5.